

What I built

- Binary classifier: given a full plasma shot, predict disruptive or stable
- Built a transformer from scratch in PyTorch — MHSA, FF blocks, encoder blocks, CLS token head
- Input: variable-length sequence of 42 plasma diagnostic features (p_rad, q95, ip_error, v_loop, etc.)
- Data: 9,760 shots from Alcator C-Mod (MIT), real experimental tokamak data

Bugs fixed along the way

- NaN loss — caused by -inf in attention mask on fully-padded rows; fixed with -1e9
- Padding mask never passed to model — padded positions were contaminating attention scores
- Fake accuracy (97%) — model predicting all-stable on 40:1 imbalanced dataset; fixed with pos_weight and F1 metric
- Data normalization leakage — was normalizing on full dataset; fixed to train stats only
- Chronological split — switched from random 80/20 to split by shot number to prevent temporal leakage
- Added gradient clipping (max_norm=1.0), lowered lr from 1e-3 to 1e-4

Results (val set, chronological split)

- F1: 0.909 | Recall: 0.957 | Precision: 0.865 on disruptive class
- Confusion matrix: 45 TP, 2 FN, 1898 TN, 7 FP out of 1952 val shots
- ROC-AUC: 0.9998 | Average Precision: 0.9920
- 1,638,913 parameters | trained in ~2 minutes on NVIDIA A40

Investigation — why so good?

- Checked for label leakage: disruptive column excluded, highest feature correlation was p_rad at 0.28
- Random forest on last timestep only: F1 = 0.25 — model is doing real work
- Random forest on full flattened sequence: F1 = 0.978 — temporal structure not necessary
- Random forest on shuffled timesteps: F1 = 0.95 — time ordering barely matters
- Conclusion: disruptive shots have different feature distributions throughout the whole shot, not just at the end
- Task is shot-level classification after the fact, not real-time early warning prediction

Early warning test

- Tested model on truncated sequences (first X% of shot visible) without retraining
- Recall flat at 0.957 from 100% down to 50% cutoff — F1 only drops below 25%
- Average shot duration ~1s, so 50% cutoff ~ 500ms warning time with no recall loss
- Disruption signature is present early in shot, not just in final moments

Limitations / what this isn't

- Not real-time prediction — feeds entire shot in at once, outputs one label
- Shot-level classification is the easy version; harder problem is time-slice prediction
- ~20 features dropped due to NaN removal (may include locked mode amplitude)
- Cross-machine generalization untested